

Command-Line Administration Guide

Essential Linux CLI Commands for System Navigators & Administrators

Navigating and managing a system via the Command Line Interface (CLI) is a foundational skill for any administrator. This reference guide covers essential commands for navigation, file management, permissions, system monitoring, and software distribution.

1. Navigation & Directory Management

These commands allow you to find your place in the hierarchical filesystem, view directory contents, transition between environments, and organize folders.

`pwd` — Print Working Directory

Purpose: Displays the absolute, full path of the directory you are currently working in from the root (`/`) directory.

```
pwd
# Output: /home/username/documents
```

`ls` — List Directory Contents

Purpose: Lists the files, symlinks, and folders contained within the specified or current working directory.

Common Flags:

- `-l` : Long listing format (displays file types, permissions, owners, size, and modified date).
- `-a` : Includes hidden files (those starting with a dot `.`).
- `-h` : Human-readable format for sizes (e.g., 4K, 25M, 2G).

```
ls -lah
```

cd — Change Directory

Purpose: Moves your current terminal context from your existing directory to a designated target location.

Common Shortcuts:

- `cd ~` or simply `cd` : Navigates straight to the user's home directory.
- `cd ..` : Moves up exactly one level into the parent directory.
- `cd -` : Returns to the previous working directory context.

```
cd /var/log
```

mkdir — Make Directory

Purpose: Generates a new, empty folder at the specified target path.

Useful Flags: `-p` creates parent directories on demand without throwing errors if they don't exist.

```
mkdir backups  
mkdir -p /var/www/html/assets/img
```

2. File Creation & Permissions Management

System administrators frequently handle configuration deployments, text updates, and script permission restrictions.

touch — Create Empty File / Update Timestamp

Purpose: Instantly initializes a blank file, or safely bumps the access/modification timestamps of an existing target to the current server time.

```
touch config.txt
```

chmod — Change File Mode / Permissions

Purpose: Modifies access constraints using absolute octal notation or symbolic flags for the Owner, Group, and Public/Others.

Octal Context: **7** (Read/Write/Execute), **6** (Read/Write), **5** (Read/Execute), **4** (Read-Only).

```
# Gives owner full access, while group & others get read/execute
chmod 755 script.sh
```

chown — Change File Owner and Group

Purpose: Adjusts individual user or group ownership mapping over administrative files. Typically requires root privileges via **sudo**.

```
# Formats as chown [owner]:[group] [target]
sudo chown admin:www-data config.txt
```

3. System Monitoring & Process Management

Administrative visibility into system resources ensures background daemons execute smoothly and unexpected spikes are managed.

ps — Process Status Snapshot

Purpose: Outputs a static list of running active processes in the system execution frame.

Common Flags: **aux** (Displays all system processes belonging to all active users, alongside specific terminal links).

```
ps aux
```

top — Dynamic Table of Processes

Purpose: Launches an interactive, real-time, scrolling visualization dashboard detailing core system load, CPU overhead, RAM allocation, and task sorting.

Interactivity: Press `q` to gracefully exit the monitoring dashboard interface.

```
top
```

4. Package Management

A core administrative duty is updating system utilities and maintaining modern service distributions.

apt install — Advanced Package Tool Package Installation

Purpose: Downloads and provisions software binaries along with their corresponding runtime dependencies on Debian/Ubuntu systems.

```
# Fetch fresh catalog tables from mirrors before installing packages
sudo apt update && sudo apt install curl -y
```